Excel Notes: Formulas, Cleaning, Charts and Pivot Tables

Study notes for Hayat | Examples use the Superstore sales dataset (columns: Order ID, Order Date, Region, Segment, Category, Product Name, Sales, Quantity, Discount, Profit).

0. Before you start: formula basics

- Every formula starts with `=`.
- **Cell references**: `A1` is relative (changes when copied), `$A$1` is absolute (never changes), `$A1` or `A$1` are mixed. Press **F4** while editing to cycle through them.
- **Ranges**: `A1:A10` is a block of cells. `A:A` is a whole column.
- **Operators**: `+ - * / ^`, comparison `= <> > < >= <=`, text join `&`.
- **Order of operations**: brackets, then `^`, then `* /`, then `+ -`.
- **Autofill**: drag the small square at the corner of a cell, or double-click it, to copy a formula down.
- **Named ranges / Tables**: Press **Ctrl+T** to turn data into a Table. Formulas can then use names like `tblSales[Sales]`, which grow automatically when you add rows.
- **Common errors**: `#DIV/0!` (divide by zero), `#N/A` (lookup found nothing), `#NAME?` (misspelled function), `#REF!` (deleted reference), `#VALUE!` (wrong data type).

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1. Basic formulas

1.1 SUM

Adds numbers.

- Syntax: `=SUM(number1, [number2], ...)`
- Example: `=SUM(G2:G9995)` gives total Sales.
- Shortcut: **Alt + =** inserts AutoSum.
- Ignores text and blank cells.
- Related: `=SUBTOTAL(109, G2:G9995)` sums only the **visible** rows after a filter (see section 4).

1.2 AVERAGE and MEAN

The arithmetic mean: sum divided by count of numbers.

- Syntax: `=AVERAGE(number1, [number2], ...)`
- Example: `=AVERAGE(J2:J9995)` gives average Profit per line item.
- **Important:** Excel has **no function called MEAN**. In statistics, mean = average, so you use `AVERAGE`. Typing `=MEAN(...)` gives `#NAME?`.
- Ignores blanks and text, but **counts zeros**. A blank is not the same as 0.
- Related functions:
  - `AVERAGEIF(range, criteria, [average_range])`: average with one condition.
  - `AVERAGEIFS(average_range, range1, crit1, ...)`: average with several conditions.
  - `TRIMMEAN(array, percent)`: mean after cutting extreme values.
  - `AVERAGEA`: also counts text as 0 and TRUE as 1 (rarely needed).

1.3 MEDIAN

The middle value when numbers are sorted. If the count is even, it averages the two middle values.

- Syntax: `=MEDIAN(number1, [number2], ...)`
- Example: `=MEDIAN(G2:G9995)`.
- **Why it matters:** the median is not pulled by outliers. If AVERAGE is much higher than MEDIAN, a few very large orders are inflating the average (the data is right-skewed).
- Quick worked example: values 2, 3, 100. AVERAGE = 35, MEDIAN = 3.
- Related: `MODE.SNGL` (most frequent value), `QUARTILE.INC`, `PERCENTILE.INC`.

1.4 SUMPRODUCT

Multiplies matching items in two or more ranges, then adds the results. It is also a flexible conditional-calculation tool.

- Syntax: `=SUMPRODUCT(array1, [array2], ...)`
- All arrays must be the same size.
- Example 1, weighted total: `=SUMPRODUCT(E2:E100, F2:F100)` where E = Quantity and F = Unit price gives total revenue.

- Example 2, conditional sum (multiply by a TRUE/FALSE test): `=SUMPRODUCT((M2:M9995="West")*(G2:G9995))` gives Sales for the West region.
- Example 3, count with two conditions: `=SUMPRODUCT((M2:M9995="West")*(O2:O9995="Technology"))`.
- Example 4, weighted average: `=SUMPRODUCT(scores, weights)/SUM(weights)`.
- Tip: `--(A1:A10>5)` turns TRUE/FALSE into 1/0. Multiplying with `*` does the same.
- Use it when `SUMIFS` can't express your logic (for example OR conditions or calculations between columns).

1.5 CONCATENATE (and its modern replacements)

Joins text from several cells.

- Old syntax: `=CONCATENATE(text1, text2, ...)`
- Example: `=CONCATENATE(B2, " - ", C2)` gives `CA-2016-152156 - Consumer`.
- Shorter way, the `&` operator: `=B2 & " - " & C2`.
- Newer functions (Excel 2019 and Microsoft 365):
  - `=CONCAT(A2:C2)` joins a whole range.
  - `=TEXTJOIN(", ", TRUE, A2:E2)` joins with a delimiter and can skip blanks (`TRUE` = ignore empty).
- Numbers and dates inside joined text need `TEXT`: `="Sales: " & TEXT(G2, "$#,##0.00")` or `=TEXT(C2, "mmm yyyy")`.
- Related text tools: `LEFT`, `RIGHT`, `MID`, `LEN`, `TRIM`, `UPPER`, `LOWER`, `PROPER`, `SUBSTITUTE`, `FIND`.

Quick reference: basic formulas

Function	What it does	Example
`SUM`	Adds numbers	`=SUM(G2:G100)`
`AVERAGE`	Mean value (Excel has no `MEAN`)	`=AVERAGE(G2:G100)`
`MEDIAN`	Middle value	`=MEDIAN(G2:G100)`
`SUMPRODUCT`	Multiply pairs, then add	`=SUMPRODUCT(S2:S100, G2:G100)`
`CONCATENATE`	Join text	`=CONCATENATE(A2, " ", B2)`

Function	What it does	Example
`MIN` / `MAX`	Smallest / largest	`=MAX(G2:G100)`
`COUNT` / `COUNTA`	Count numbers / count non-empty	`=COUNTA(A2:A100)`
`ROUND`	Round to n decimals	`=ROUND(G2, 1)`

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2. Advanced formulas

2.1 IF

Returns one value if a condition is true and another if it is false.

- Syntax: `=IF(logical_test, value_if_true, value_if_false)`
- Example: `=IF(J2<0, "Loss", "Profit")`.
- **Nested IF** (several outcomes): `=IF(G2>=1000, "High", IF(G2>=200, "Medium", "Low"))`. Excel checks left to right and stops at the first true test.
- Combine conditions with `AND(...)` and `OR(...)`:
  - `=IF(AND(M2="West", G2>500), "Priority", "Normal")`
  - `=IF(OR(M2="East", M2="West"), "Coast", "Inland")`
- Newer alternative to deep nesting: `=IFS(G2>=1000,"High", G2>=200,"Medium", TRUE,"Low")`.
- **IFERROR** hides errors: `=IFERROR(J2/G2, 0)` returns 0 instead of `#DIV/0!`. Use it carefully; do not hide real mistakes.

2.2 COUNTIF and COUNTIFS

Counts cells that meet a condition.

- Syntax: `=COUNTIF(range, criteria)`
- Examples:
  - `=COUNTIF(M2:M9995, "West")` counts rows in the West region.
  - `=COUNTIF(J2:J9995, "<0")` counts loss-making rows.
  - `=COUNTIF(G2:G9995, ">"&A1)` uses a value from another cell (join the operator with `&`).

- **Wildcards** (text only): `*` means any characters, `?` means one character. `=COUNTIF(Q2:Q9995, "*Chair*")` counts product names containing "Chair".
- Multiple conditions: `=COUNTIFS(M2:M9995, "West", O2:O9995, "Technology")`. All conditions must be true (AND logic).
- Count blanks: `=COUNTBLANK(range)`.

2.3 SUMIF and SUMIFS

Adds numbers that meet a condition.

- Syntax: `=SUMIF(range, criteria, [sum_range])`
- Example: `=SUMIF(M2:M9995, "West", G2:G9995)` gives total West Sales.
- **SUMIFS** takes the sum range **first**, then pairs of range and criteria: `=SUMIFS(G2:G9995, M2:M9995, "West", O2:O9995, "Technology")`
- Dates: `=SUMIFS(G2:G9995, C2:C9995, ">="&DATE(2016,1,1), C2:C9995, "<="&DATE(2016,12,31))`.
- Criteria pointing at a cell: `=SUMIFS(G:G, M:M, $A2)` lets you drag the formula down a list of regions.
- Remember: in `SUMIF` the sum range is last; in `SUMIFS` it is first.

2.4 VLOOKUP

Looks up a value in the **first column** of a table and returns a value from a column to the right.

- Syntax: `=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`
- Example: `=VLOOKUP("FUR-BO-10001798", A2:C500, 3, FALSE)` returns the value from the 3rd column of the table.
- **Always use `FALSE` (or 0) for exact match.** `TRUE` (approximate match) needs sorted data and is only for things like grade bands or tax brackets.
- Limitations:
  - Cannot look to the **left** of the lookup column.
  - Breaks if you insert a column (the column number stays the same).
  - Slower on big tables.
- Lock the table with `$`: `=VLOOKUP(A2, $F$2:$H$100, 2, FALSE)`.
- Wrap in `IFERROR` for clean output: `=IFERROR(VLOOKUP(...), "Not found")`.

2.5 INDEX and MATCH

A more flexible lookup pair.

- `=MATCH(lookup_value, lookup_array, [match_type])` returns the **position** of a value. Use `0` for exact match.
- `=INDEX(array, row_num, [column_num])` returns the value at a position.
- Combined: `=INDEX(return_range, MATCH(lookup_value, lookup_range, 0))`
- Example: `=INDEX(Q2:Q9995, MATCH("CA-2016-152156", B2:B9995, 0))` returns the product name for that order.
- Advantages over VLOOKUP: looks in **any direction**, survives inserted columns, and is faster.
- Two-way lookup (row and column): `=INDEX(B2:F10, MATCH("West", A2:A10, 0), MATCH("Q3", B1:F1, 0))`.
- Modern alternative (Excel 2021 and Microsoft 365): `=XLOOKUP(lookup_value, lookup_array, return_array, "Not found")`. It replaces both VLOOKUP and INDEX/MATCH for most jobs. Older Excel versions do not have it.

Quick comparison: lookup methods

Feature	VLOOKUP	INDEX + MATCH	XLOOKUP
Looks left	No	Yes	Yes
Survives inserted columns	No	Yes	Yes
Default match	Approximate (risky)	Set by you	Exact
Available in	All versions	All versions	2021 / 365

Quick reference: conditional functions

Goal	Formula
Count rows where Region is West	`=COUNTIF(M:M, "West")`
Sum Sales for West	`=SUMIF(M:M, "West", G:G)`
Sum Sales for West and Technology	`=SUMIFS(G:G, M:M, "West", O:O, "Technology")`

Goal	Formula
Average Profit for Consumer	`=AVERAGEIF(H:H, "Consumer", J:J)`
Label each row	`=IF(J2<0, "Loss", "Profit")`
Safe division	`=IFERROR(J2/G2, 0)`

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3. Remove duplicates and conditional formatting

3.1 Find duplicates before deleting

1. **Highlight them first**: select the column, then **Home > Conditional Formatting > Highlight Cells Rules > Duplicate Values**.
2. **Flag with a formula**: `=COUNTIF($A$2:$A$100, A2)>1` returns TRUE for repeated values.
3. **Flag only the 2nd+ copy**: `=COUNTIF($A$2:A2, A2)>1`. The first occurrence stays FALSE.

3.2 Remove Duplicates tool

1. Click inside your data (or select it).
2. **Data > Remove Duplicates**.
3. Tick the columns that define a duplicate. Ticking all columns means only rows that are identical in every column are removed. Ticking only `Order ID` removes all but the first row of each order, which is **not** what you want for line-item data.
4. Tick **My data has headers**, then click OK. Excel reports how many duplicates were removed and how many remain.

Safety tips

- Work on a **copy** of the sheet. Remove Duplicates permanently deletes rows and keeps the **first** occurrence.
- Remove stray spaces first with `=TRIM(A2)`, otherwise "West" and "West " count as different values.
- Text comparison in Excel is **not case-sensitive** ("west" = "West").

Other methods

- **Advanced Filter**: Data > Advanced, tick **Unique records only**, copy to another location.
- **Power Query**: Data > Get Data > From Table/Range, then Home > Remove Rows > Remove Duplicates. It records the step so it repeats on refresh.
- **UNIQUE function** (Excel 2021 and 365): `=UNIQUE(A2:A100)` spills a list of distinct values without changing the source.

3.3 Conditional formatting

Formats cells automatically based on their values. Found at **Home > Conditional Formatting**.

Type	Use it for
Highlight Cells Rules	Greater than, less than, between, equal to, text contains, duplicates
Top/Bottom Rules	Top 10 items, top 10%, above/below average
Data Bars	Small in-cell bars showing size
Color Scales	Heat map (for example red to green)
Icon Sets	Arrows, traffic lights, flags
New Rule > formula	Anything custom

Formula-based rule (very powerful)

1. Select the range (for example `A2:J100`).
2. **New Rule > Use a formula to determine which cells to format**.
3. Enter a formula written for the **top-left cell** of the selection, with `$` locking the column: `=$J2<0`
4. Choose a format (red fill) and click OK. The **entire row** turns red when Profit in column J is negative.

More examples:

- Highlight rows from the West region: `=$M2="West"`
- Highlight dates in the past: `=$C2<TODAY()`
- Highlight every other row: `=MOD(ROW(),2)=0`

Managing rules: **Conditional Formatting > Manage Rules** lets you edit, reorder, set *Stop If True*, and change "Applies to". **Clear Rules** removes them.

Good practice: use few colors, keep meaning consistent (red = bad, green = good), and always add a legend if the dashboard is shared.

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4. Charts, filters, sort and slicers

4.1 Sort

- **Data > Sort A to Z / Z to A** for one column. Click inside the data first so Excel sorts the whole table, not one column.
- **Custom sort**: Data > Sort lets you add levels, for example Region (A to Z), then Sales (Largest to Smallest).
- You can sort by cell color, font color or icon.
- **Custom list** (Jan, Feb, Mar... or Low, Medium, High): in the Sort dialog choose Order > Custom List.
- Warning: if formulas use relative references to other rows, sorting can change results. Keep a Row ID column so you can restore the original order.

4.2 Filter

- Turn on with **Ctrl + Shift + L** or Data > Filter. Dropdown arrows appear in the header row.
- Filters can be text (contains, begins with), number (greater than, top 10, above average) or date (this month, last quarter).
- Filters across several columns combine with AND.
- Filtered-out rows are **hidden, not deleted**.
- Use `SUBTOTAL(109, range)` to total only the visible rows.
- **Advanced Filter** (Data > Advanced) lets you filter with a separate criteria range, including OR logic, and copy results elsewhere.
- **FILTER function** (Excel 2021 and 365): `=FILTER(A2:J100, M2:M100="West", "No data")` returns a live list.

4.3 Charts

Select your data (with headers), then **Insert > Recommended Charts**, or pick a type directly.

Goal	Best chart
Trend over time	Line (or column for few periods)
Compare categories	Column or bar (use bar for long labels)
Parts of a whole	Doughnut or pie (only for a few categories, 5 or fewer)
Relationship between two numbers	Scatter
Ranking	Sorted bar chart
Two measures with different scales	Combo chart with a secondary axis
Distribution	Histogram

Building a good chart

1. Add a clear **title** that tells the message, for example "Technology earns the most profit".
2. Label axes and use a number format like `$#,##0,"K"` (shows 125,000 as $125K).
3. Remove clutter: heavy gridlines, 3D effects, unnecessary legends.
4. Add **data labels** only where they help.
5. Sort bar charts from largest to smallest.
6. Keep colors consistent across charts.

Chart tools: Chart Design tab (change type, switch rows/columns, select data) and Format tab. The **+** button next to the chart toggles chart elements. **Sparklines** (Insert > Sparklines) draw tiny charts inside a single cell.

Tip: charts based on a Table or PivotTable update automatically when the data changes.

4.4 Slicers

Visual, clickable filter buttons.

1. Click inside a **Table** or **PivotTable**.
2. **Insert > Slicer** (or PivotTable Analyze > Insert Slicer).

3. Tick the fields (for example Region, Segment, Year).
4. Click the buttons to filter. Hold **Ctrl** to select several, or use the funnel icon with an X to clear.
5. **Timeline** (Insert > Timeline) is a slicer for dates (months, quarters, years).
6. **Connect one slicer to several PivotTables**: select the slicer, then Slicer tab > **Report Connections** and tick all PivotTables. This is how a dashboard filters every chart at once.
7. Style and button columns can be changed on the Slicer tab.

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5. Pivot tables and pivot charts

5.1 What a PivotTable is

A tool that summarizes thousands of rows into a small table by dragging fields. It does not change your source data and needs no formulas.

Prepare the data first

- One header row, unique and non-blank headers.
- No blank rows or columns, no merged cells.
- One type of data per column (dates as real dates, numbers as numbers).
- Best practice: convert to a **Table** (Ctrl + T) so new rows are included automatically.

5.2 Create a PivotTable

1. Click any cell in the data.
2. **Insert > PivotTable**. Choose the table/range and a location (new worksheet is recommended).
3. In the **PivotTable Fields** pane, drag fields into four areas:

Area	Purpose	Example
Rows	Categories listed down the side	Category, Region, Product
Columns	Categories listed across the top	Year, Segment
Values	Numbers to calculate	Sum of Sales, Sum of Profit
Filters	A report-wide filter	Region

Example: Rows = Category, Columns = Year, Values = Sum of Sales gives total sales per category per year.

5.3 Values settings

- Right-click a value > **Summarize Values By**: Sum, Count, Average, Max, Min, and more.
- **Show Values As**: % of Grand Total, % of Column Total, % of Row Total, Running Total, Difference From (for example month-over-month).
- **Number Format**: right-click a value > Number Format to apply currency or percentages.
- Count of Order ID counts line items. For **distinct orders**, tick **Add this data to the Data Model** when creating the PivotTable, then use **Distinct Count**.

5.4 Grouping

- **Dates**: right-click a date in the pivot > Group > choose Months, Quarters and Years. You get Year and Month fields without helper columns. (Excel 2016 and later group dates automatically.)
- **Numbers**: right-click > Group, set start, end and interval (for example Discount bands of 0.1).
- **Text**: select items, right-click > Group to make custom groups.

5.5 Sort, filter and format inside a PivotTable

- Sort by value: right-click a value > Sort > Largest to Smallest.
- **Value filters / Top 10**: row label dropdown > Value Filters > Top 10 (use this for "top 10 products").
- **Label filters** and search box for text.

- Design tab: Report Layout (Compact, Outline, Tabular), Subtotals, Grand Totals, Banded Rows, styles.
- Options: tick **Autofit column widths on update** off if you want your widths kept; set **For empty cells show: 0**.

5.6 Calculated fields

PivotTable Analyze > Fields, Items & Sets > **Calculated Field**. Example: Name `Margin`, Formula `=Profit/Sales`. Note it divides the **sums**, which is what you usually want for margin.

5.7 Refresh

PivotTables do **not** update automatically when the source changes. Use **Right-click > Refresh** or **Alt + F5** (Ctrl + Alt + F5 refreshes all). If the source is a Table, new rows are picked up on refresh. You can also set Options > Data > **Refresh data when opening the file**.

5.8 GETPIVOTDATA

When you click a pivot cell from a formula, Excel writes `=GETPIVOTDATA(...)`. It stays correct even if the pivot layout changes. Turn it off with PivotTable Analyze > Options > uncheck **Generate GetPivotData**.

5.9 PivotCharts

1. Click inside the PivotTable, then **Insert > PivotChart** (or PivotTable Analyze > PivotChart).
2. Choose a chart type (not every type works, for example scatter is unavailable).
3. The chart is **linked** to the pivot: filtering the pivot, using a slicer, or changing fields changes the chart. Field buttons on the chart act as filters. You can hide them from PivotChart Analyze > Field Buttons.
4. Format it like any chart (title, colors, data labels).

5.10 Build an interactive dashboard (workflow)

1. Clean the data (remove duplicates, fix dates and text) and make it a Table.
2. Create one PivotTable per view: monthly sales, profit by category, sales by region, top 10 products.

3. Insert a PivotChart for each.
4. Insert slicers (Year, Region, Segment) and **connect them to every PivotTable** via Report Connections.
5. Put the charts and a few KPI cards on one sheet, hide gridlines (View > uncheck Gridlines), and line up the objects.
6. Test: click each slicer and check every chart responds.

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6. Practice exercises (Superstore data)

1. Find total Sales, average Profit, and the median Sales value. Compare average and median. What does the gap tell you?
2. Use `SUMPRODUCT` to calculate total Sales for Technology orders in the East region. Check it against `SUMIFS`.
3. Create a column that joins Customer Name and City using `&`, and again using `TEXTJOIN`.
4. Label each row "Loss" or "Profit" with `IF`, then count each label with `COUNTIF`.
5. Build a small table of Regions with `SUMIFS` for Sales and Profit and a margin column with `IFERROR`.
6. Look up the product name for a given Product ID using `VLOOKUP`, then redo it with `INDEX` and `MATCH`.
7. Highlight duplicate Order IDs with conditional formatting, then explain why they are not real duplicates (an order has several line items).
8. Apply a color scale to the margin column and data bars to Sales.
9. Sort by Region, then Profit (largest first), and filter to loss-making rows only.
10. Create a PivotTable: Rows = Category, Columns = Year, Values = Sum of Profit. Add a PivotChart, a Region slicer, and a Date timeline.

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7. Shortcut cheat sheet

Action	Shortcut
AutoSum	Alt + =
Toggle absolute/relative reference	F4

Action	Shortcut
Create a Table	Ctrl + T
Toggle filter	Ctrl + Shift + L
Select current region	Ctrl + A (inside data)
Copy formula down	Ctrl + D
Insert a chart from selection	Alt + F1
Refresh PivotTable	Alt + F5
Show formulas	Ctrl + `
Fill with Ctrl + Enter	Type a formula, press Ctrl + Enter to fill the selection

8. Common mistakes to avoid

- Using `VLOOKUP` without `FALSE` for exact match.
- Forgetting `$` so a lookup range shifts as you copy the formula down.
- Numbers or dates stored as text (they left-align and won't sum). Fix with Text to Columns or `VALUE`.
- Extra spaces causing failed matches. Use `TRIM`.
- Removing duplicates on the wrong columns.
- Not refreshing a PivotTable after the data changes.
- Hiding errors with `IFERROR` instead of fixing the cause.
- Typing `=MEAN()`: use `AVERAGE`.